# Observables as Projections of a $\pi$-free Graph: A Synthesis of the GeoVac Precision-Observation Arc*

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The Geometric Vacuum (GeoVac) framework separates into two structurally distinct layers, and that separation is the organizing principle of its precision-observation arc. *Layer 1* is a bare combinatorial graph: quantum-number labels $(n, \ell, m, \ldots)$, rational matrix entries, algebraic eigenvalues, no physical units, and—verified symbolically—no $\pi$. *Layer 2* is a finite family of named projections from that graph onto continuous physical settings; every physical variable, every dimension, and every transcendental constant that appears in any GeoVac calculation enters through exactly one projection and is traceable to it. This synthesis assembles the four Group 6 papers into a single narrative built on that thesis. The spine is: the projection taxonomy and its three-axis tagging (Paper 34 [11]); time as the prototype *free-side* projection—$\pi$ enters the Klein–Gordon spectrum on $S^3 \times \mathbb{R}$ at exactly one step, temporal compactification, while rest mass preserves the algebraic ring (Paper 35 [12]); entropy as an embedding-class projection artifact, identically zero on the one-body graph and finite only after the metric embedding of the two-body repulsion (Paper 27 [7]); and the entanglement structure that isolates that repulsion in the angular-momentum eigenbasis (Paper 26 [6]). A multi-observable focal-length program closes the arc: Roothaan autopsies decompose precision atomic observables into projection-chain-tagged components, exposing literature convention mismatches and framework kernel gaps that single-observable analyses cannot see. Two disciplines run throughout. *First*, every transcendental named here is tagged to its projection—$\pi$ via the M1 volume slot (unit-sphere and Fock volumes), $\pi^{2k}$ via the spectral action, $\zeta(2k)$ via even-zeta Dirichlet series, odd $\zeta$ via the half-integer-Hurwitz spinor lift, Catalan $G$ via quarter-integer vertex parity, $2\pi$ via temporal compactification. *Second*, the precision matches are sub-percent *framework-native* agreements that surface convention and kernel issues; they are *not* first-principles predictions of the constants, and the multi-focal-composition wall—the framework has no native theorem for composing multiple Fock-style projections—is named as an active boundary. The unifying reading is that compactness forces discreteness (the converse is false: discrete spectrum is the compact-resolvent property), made operational: the discrete skeleton is the compact/closed regime, and a transcendental is precisely the price of releasing compactness by integrating over a promoted continuous parameter.

## I. INTRODUCTION: TWO LAYERS, ONE ORGANIZING OBSERVATION

The GeoVac quantum-chemistry and quantum-computing arcs (Groups 2 and 4) treat the framework as a sparse graph-Hamiltonian engine. The operator-algebra arc (Group 1) treats the same construction as a discrete spectral triple. These two readings are not in conflict; they describe two different *layers* of one object, and the precision-observation arc collected here is the arc that names the layers and the bridge between them.

Layer 1 is the bare combinatorial graph. Its nodes are representation-theoretic labels of compact Lie groups; its spectral data are integers — the Casimir eigenvalue $n^2 - 1$ that labels the Coulomb $S^3$ shells (the *continuum* Laplace–Beltrami spectrum the graph encodes as labels; the graph Laplacian itself is a different, positive-semidefinite operator) and the total quanta $N$ on the Bargmann–Segal $S^5$ graph; its matrix elements are rational or algebraic over $\mathbb{Q}[\sqrt{2}, \sqrt{3}, \sqrt{6}, \ldots]$ from Wigner $3j$ algebra. That no $\pi$ appears in Layer 1 is not an aspiration but a verified certificate: on both the Coulomb $S^3$ Fock graph and the Bargmann–Segal $S^5$ graph, every adjacency entry, every eigenvalue, and every coupling coefficient is exact rational/algebraic in exact arithmetic (Paper 24 [5], Paper 28's graph-native QED certificate). This is a SYMBOLIC certificate, not a numerical observation.

Layer 2 is a finite family of named maps from the bare graph to physical settings: the Fock 1935 conformal projection, the Hopf bundle, the Bargmann–Segal transform, the Connes–Chamseddine spectral action, the Sturmian reparameterization, the Camporesi–Higuchi spinor lift, the Wigner $3j/D$ couplings, temporal compactification, and others. Each projection introduces specific physical variables ($Z$, $\alpha$, $\hbar\omega$, $R$, $\beta$, $\Lambda$, $\ldots$), attaches specific dimensions to graph data, and carries a specific transcendental signature. The load-bearing structural claim of the arc is an Observation, asserted at exactly that tier:

**Observation 1** (Single-projection origin of physical content)**.** *Every coupling constant, energy scale, length scale, and transcendental constant appearing in any GeoVac computation traces to exactly one projection in Layer 2. The bare graph contains none of $\alpha$, $Z$, $\hbar\omega$, $R$, $\Lambda$, $\pi$, or $\zeta(2k)$; these enter by, and only by, the projection the calculation invokes.*

* Group 6 synthesis in the Geometric Vacuum series

This is the operational form of the project's organizing observation that *compactness forces discreteness* (not a biconditional: the Paper 24 oscillator has discrete spectrum on non-compact $\mathbb{R}^3$). Compact groups have discrete spectra (Peter–Weyl), and the bit-exact rational skeleton is the closed/compact regime; the continuum—and with it every transcendental—is what remains when compactness is released. Calibration data is continuum (un-packed) data, which is why the skeleton is forced and rational while the projections are free and transcendental. The four Group 6 papers make this concrete: Paper 34 catalogues the projections; Paper 35 locates the exact step at which the prototype free-side projection (time) injects $\pi$; Papers 27 and 26 show entropy and entanglement to be projection artifacts, zero on the bare graph and finite only after metric embedding.

We hold one framing discipline throughout, following the GeoVac rhetoric rule: the discrete graph and its continuous projections are dual descriptions connected by proven maps, and no ontological priority is asserted for either. The contribution is a vocabulary that makes the question "where does this $\pi$ come from?" answerable, not a claim about which description is fundamental.

## II. THE PROJECTION TAXONOMY AND MASTER-MELLIN TAGGING (PAPER 34)

Paper 34 is the keystone of this group: the living catalogue of *twenty-eight* projection mechanisms, each tagged along three independent axes—(a) the physical variables it introduces, (b) the dimension or unit it attaches to graph data, and (c) its transcendental signature in the sense of Paper 18 [4]. The transcendental axis is a structural sibling of Paper 18's transcendental taxonomy: each tier corresponds to a projection type, and reading the dictionary in either direction recovers the other. A representative slice of the dictionary is reproduced in Table I.

*a. Where the transcendentals live.* The dictionary makes the tagging discipline mechanical. The kinetic constant $\kappa = -1/16$ and the volume $\pi$ enter at the Fock conformal projection; the dimensionless $\pi = \mathrm{Vol}(S^2)/4$ at the Hopf bundle; the $\pi^{2k} \cdot \mathbb{Q}$ content of vacuum-polarization-type observables at the Connes–Chamseddine spectral action ($\Pi = 1/(48\pi^2)$ itself is the entered flat-space value, Paper 28); the even-zeta value $\zeta_R(2)$ (the Fock-Dirichlet object $F$) at a Dirichlet series at $s = 4$; the half-/quarter-integer Hurwitz content that promotes to odd $\zeta$ (unrestricted $|D|$ sums at odd $s$) and, at the Camporesi–Higuchi spinor lift's vertex selection, Catalan's $G$ and $\beta(4)$; and $2\pi \cdot \mathbb{Q}$ at temporal compactification (Sec. III). None of these is an anonymous transcendental: each is pinned to a projection row, which is the discipline this group exists to model. The per-loop $S^2$ Weyl constant $1/(4\pi)$ (M1 volume slot) is tagged to the vector-photon promotion; the Paper 2 [2] combination rule $K = \pi(B + F - \Delta)$ is a separate object (it does not consume the per-loop $1/(4\pi)$) and remains an

| Projection | Dimension | Transcendental signature |
|---|---|---|
| Fock conformal | energy | $\kappa = -1/16$; $\pi$ via $\mathrm{Vol}(S^3)$ |
| Hopf bundle | dimensionless | $\pi = \mathrm{Vol}(S^2)/4$ |
| Bargmann–Segal | energy | *none* (graph $\pi$-free) |
| Spectral action | energy | $\pi^{2k} \cdot \mathbb{Q}$ observables |
| Spinor lift (CH) | dimensionless | half-int. Hurwitz $\to$ odd $\zeta$; with vertex parity, $G$, $\beta(4)$ |
| Wigner $3j$ | dimensionless | $\mathbb{Q}[\sqrt{2k+1}]_k$ |
| Vector-photon | dimensionless | $1/(4\pi)$ per loop |
| Rest-mass | mass | *trivial* (ring-preserving) |
| Temporal window | time | $2\pi \cdot \mathbb{Q}$ per Matsubara mode |

TABLE I. A slice of the Paper 34 projection dictionary. Each row is a named structural construction that introduces specific physical content into Layer 1 graph data; the transcendental signature is in the sense of Paper 18. The full dictionary catalogues twenty-eight projections.

**Observation**, never a derivation, and the taxonomy is consistent with that tiering rather than strengthening it.

*b. The engine versus the catalogue.* The computational engine behind the transcendental tags is the master Mellin engine—the Mellin transform of $\mathrm{Tr}\,(D^k e^{-tD^2})$ at $k \in \{0, 1, 2\}$, whose three domains $M_1/M_2/M_3$ are read as generating the volume-slot $\pi$-signatures (e.g. the convention-dependent unit-sphere volume quotient $4/\pi$, its M1 placement a reading), the pure-Tate $\pi^{2k}$ tier at integer $s$ (the odd zeta values at half-integer $s$ are M3's, Paper 18), and the vertex-parity Catalan/$\beta$ tier respectively (slot assignments and ring homes: Paper 18 §III.7 and Paper 55, the single owners). That engine is developed in Paper 18 §III.7 (Group 3, the foundations trunk); Paper 34 is the *catalogue* that consumes its output and assigns each tier to a projection. The synthesis-level reading is that the transcendental axis is the only genuinely independent axis of the three: Sprint TX-A's[1] dimensional axiomatization found the variable and dimension axes to be perfectly correlated, so the dictionary's content reduces to the projection $\leftrightarrow$ transcendental map.

## III. TIME AS THE PROTOTYPE FREE-SIDE PROJECTION (PAPER 35)

Paper 35 answers the question Paper 34 leaves open—*where exactly* does $\pi$ enter, and at no other step—in the simplest non-trivial relativistic system the framework describes: a free scalar on $S^3 \times \mathbb{R}$ governed by the Klein–Gordon equation.

[1] Sprint designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

*a. The before/after.* The bare Klein–Gordon spectrum is $\pi$-free in the algebraic-extension ring $\mathbb{Q}[\sqrt{d_i}]$ for every rational mass-squared parameter. This is verified SYMBOLICALLY for $n \in [1, 50]$ across the panel $m^2 \in \{0, 1, 1/4, 2\}$ (200 cases, zero transcendentals). Mass is therefore a *rest-mass projection*: it introduces a parameter of dimension mass and *preserves* the bare-graph ring. Compactify the temporal direction, $S^3 \times \mathbb{R} \to S^3 \times S^1_\beta$, and the spectrum acquires Matsubara modes $\omega_k^t = 2\pi k/\beta$; the first $\pi$-bearing eigenvalue is the $(n{=}0, k{=}1)$ mode with $\omega^2 = 4\pi^2$. Every $(n, 0)$ eigenvalue is $\pi$-free; every $(n, k \neq 0)$ eigenvalue carries $2\pi/\beta$. The conformally coupled massless scalar Casimir energy on unit $S^3$ (spatial only, no temporal compactification) is the exact rational $E_{\text{Cas}} = 1/240$, with no transcendental at any step; the Stefan–Boltzmann constant $\pi^2/90$ appears only in the high-temperature expansion on $S^3 \times S^1_\beta$—through the Matsubara sum, i.e. exactly the spectral $\pi$-injection.

*b. The sharpened reading and its tier.* This forces a refinement of the Paper 34 dictionary: the implicit "scalar parameter projection" splits into a ring-preserving *rest-mass projection* (introduces $m$, signature trivial) and an *observation/temporal-window projection* (introduces $\beta$, signature $2\pi \cdot \mathbb{Q}$ per mode). The structural statement is that a GeoVac observable contains $\pi$ if and only if its evaluation includes a continuous integration over a temporal or spectral parameter promoted from the discrete graph spectrum. This is a structural OBSERVATION supported by a symbolic verification panel (the 208-case panel that Paper 34's prediction subsection cites as a downstream control); it is asserted at that tier, not as a theorem about all observables. Time is, in this reading, the prototype free-side projection: the observer's finite-time measurement window, read as a periodic Euclidean circle (an identification Paper 35 adopts rather than derives; a compact interval without periodic identification keeps a continuous spectrum, and whether a real-time window forces the circle is open), is the formal mechanism by which the continuum—and $\pi$—enters.

This connects directly to the project's internal working hypothesis that time-discreteness is observer-compactification (WH7): the only temporal structure the metric can see is compactified time, and compactified time, read as a circle, has a discrete spectrum. We flag the honest caps that internal diagnostic work attaches to that hypothesis—a real-time window becomes a circle only through the identification above, the Matsubara circle is Euclidean while an observer's window is real-time (Paper 35 cites the published bound on that reading), and whether the Lorentzian *signature* is metrically visible at finite cutoff remains open, and the relevant falsifier (a metrically visible temporal algebra on a non-compact carrier) is an active probe, not a closed result. Paper 35's load-bearing content—that $\pi$ enters at temporal compactification and that rest mass is ring-preserving—stands independently of that open question.

## IV. ENTROPY AS AN EMBEDDING-CLASS PROJECTION (PAPER 27)

Paper 27 places entanglement entropy in the same two-layer frame and shows it to be a projection artifact rather than an intrinsic feature of the graph.

*a. One-body operators are entanglement-inert.* The operator-theoretic result is an INTERNAL THEOREM: on any finite lattice, the von Neumann entropy of any reduced density matrix derived from a *non-degenerate* ground state of a purely one-body fermionic Hamiltonian is identically zero (the ground state is a single Slater determinant). Numerically this is confirmed to machine precision, $S_{\text{kin}}/S_{\text{full}} \sim 10^{-14}$ for He at $n_{\max} = 2, 3$. All ground-state entanglement is generated by the two-body interaction $V_{ee}$. The non-degeneracy qualifier is essential—open-shell and dissociated-bond states escape the floor—and is stated as such.

*b. The "area law" is pair counting.* Earlier framework material presented a holographic scaling $S_n = k \log A_n$ with $A_n \propto n^4$ in one-particle language. Paper 27 re-derives it rigorously as a *pair-counting* statement: $A_n = g_n^2$ with shell degeneracy $g_n = 2n^2$, so the factor of four is the two-body signature, not a genuine area law. This corrects the earlier presentation in place rather than propagating it—a model of the two-way verdict discipline (a claim is re-priced to what its backing supports).

*c. A second rigidity statement* — RETRACTED. An earlier version of this synthesis reported, at INTERNAL THEOREM tier, that the two-fermion closed-shell ground-state entropy on the Bargmann–Segal harmonic-oscillator lattice is identically zero for any central two-body potential, via Moshinsky–Talmi total-quanta conservation of $N_1 + N_2 = N_{\text{rel}} + N_{CM}$. **That claim, its mechanism, and its tier are withdrawn** (Papers 24 and 27, retracted 2026-08-22). The Moshinsky–Talmi *bracket* conserves total quanta, but the *matrix element* of a central $V(r_{12})$ does not: it couples different relative-$n$ (for the Minnesota singlet the off-diagonal $\langle n{=}0|V|n{=}1\rangle = +17.3$ MeV is $\sim 31\times$ the $-0.55$ MeV diagonal). Corrected, the entropy is small but *nonzero* and *grows* with basis size (0.0671/0.0716/0.0833 nats at $N_{\max} = 2/3/4$), the ground state is *not* a single Slater determinant, and $\|[H_{HO}, V]\|_F / \|H_{HO}\|_F$ is $O(1)$ (0.74/0.63/0.67), not at the noise floor — the published $\sim 10^{-16}$ was an undocumented-guard artifact. So there is no entropy-side rigidity dual: Paper 24's HO rigidity theorem (the spectral, structural parent statement) stands, but this two-fermion entropy corollary does not.

*d. Taxonomy placement.* Entropy slots cleanly as an *embedding-class* observable in the Paper 18 taxonomy and a projection artifact in the Paper 34 frame: zero on the combinatorial graph (one-body operator alone, no transcendental), finite only after the metric embedding of $1/r_{12}$ onto the pair-state graph, where the spectral eigenvalues of $V_{ee}$ become irrational. The combinatorial graph is entropy-blind; entropy is the price of projecting onto

the metric continuum. A measured refinement accompanies the placement: Coulomb-lattice block entropy obeys a universal two-variable scaling $S_B = A\,(\tilde{w}_B/\delta_B)^{\gamma}$ whose local exponent extrapolates to $\gamma_\infty \approx 1.93$–$1.96$ (the precise value is extrapolation-order dependent), in every case below the non-degenerate second-order Rayleigh–Schrödinger value of 2—a MEASURED extrapolation, reported with its residual gap attributed to multi-shell aggregation, not claimed as exactly 2.

## V. ENTANGLEMENT STRUCTURE OF THE ANGULAR-MOMENTUM BASIS (PAPER 26)

Paper 26 supplies the computational backbone for the entropy thesis by characterizing *where* the two-body entanglement lives in the angular-momentum eigenbasis—the basis of hydrogenic radial functions times spherical harmonics labeled by $(n, \ell, m)$.

*a. Energy–entanglement decoupling.* At the fixed exponent $k = Z$ the one-body Hamiltonian of a He-like ion is exactly diagonal in this basis, so the off-diagonal electron repulsion carries all of the correlation energy and all of the entanglement; the total entanglement entropy scales as $S \sim Z^{-2.52}$ across He-like ions (MEASURED, $Z = 2$–$10$ at $n_{\max} = 4$). An earlier version assigned 10–39% of the [retracted 2026-09-25: he-graph-native-hop-headline] correlation energy to an off-diagonal "graph-Laplacian" one-body term (coupling $\kappa = -1/16$) and quoted $S \sim Z^{-2.56}$; that term is a hop that is not part of the ion's Hamiltonian, so the attribution is withdrawn and the exponent re-measured without it (2026-09-25, Paper 26). Added as a model perturbation, the hop still illustrates the one-body entanglement-inertness theorem: it moves the model energy by 10–39% of the correlation energy while generating $< 0.2\%$ of [retracted 2026-09-25: he-graph-native-hop-headline] the entanglement. The structural origin is that a one-body operator is separable and cannot entangle orbitals, while $V_{ee}$ is the irreducible two-body interaction; the angular-momentum basis, by nearly diagonalizing the one-body physics, isolates the entanglement in $V_{ee}$ alone.

*b. Basis-intrinsic sparsity.* Among orbital bases for spherically symmetric systems, the angular-momentum eigenbasis is an *isolated* point of ERI sparsity: Any generic (angular-mixing) unitary rotation of orbital indices, however small, fills the ERI tensor from 42% to 100% density in a rapid transition at the identity, and the nonzero ERI count is $Z$-independent (sparsity is purely geometric). This MEASURED transition is the entanglement-side counterpart of the potential-independent angular-sparsity theorem (Paper 22, Group 3) and the structural root of the exactly $Q$-linear composed Pauli count ($N_{\text{Pauli}} = 27.90 \times Q$) reported in the quantum-computing arc (Group 4): the same basis that decouples one-body energy from two-body entanglement is the one that produces sparse ERIs.

*c. Network topology and core–valence decoupling.* Orbital mutual-information maps exhibit a hub-migration pattern ($1s \to 2s \to 2p$) tracking the partially filled shell across the first row, and core–valence mutual information falls to $\approx 2 \times 10^{-3}$ by $Z = 4$ and at the $\sim 10^{-15}$ noise floor for $Z \geq 5$ (basis saturation at $n_{\max} = 2$). The latter is consistent with the composed fiber-bundle factorization used in the molecular framework, but does not establish it: at $n_{\max} = 2$ the core is exactly closed, so the core-valence mutual information vanishes by construction (Paper 26 §V scope note). Paper 26 states its limitations plainly—$n_{\max} = 2$ for the first-row network maps, $R$-independence is a property of the composed architecture (not of the true molecular ground state, whose entanglement diverges at dissociation), and the $d$-orbital extension is open.

## VI. THE MULTI-OBSERVABLE FOCAL-LENGTH PROGRAM (PAPER 34 §V)

The arc's most distinctive instrument is the multi-observable focal-length decomposition program: the discovery that a global fit of several precision observables through one structural dictionary exposes systematics that no single-observable analysis can see.

*a. Roothaan autopsies.* For observables with a multi-component literature decomposition—the hydrogen Lamb shift, the 21 cm and muonic hyperfine intervals, the muonic-deuterium Lamb shift—Paper 34 hosts *Roothaan autopsies*: decomposition tables that present each measured observable as a sum of focal-length-tagged components, each carrying its explicit projection chain and a framework-native (FN) versus Layer-2-input (L2) status. The hydrogen $2S_{1/2}$–$2P_{1/2}$ Lamb-shift autopsy is representative: the framework-native components (self-energy via Fock $\circ$ Sturmian $\circ$ spectral action $\circ$ Drake–Swainson, Uehling vacuum polarization via Fock $\circ$ spectral action, …) sum to +1057.19 MHz, 99.9% of the 1057.845(9) MHz measurement, with the Layer-2 inputs (multi-loop QED, finite-size, recoil NLO) netting to −1.06 MHz — they do not cancel; an earlier near-cancellation reading rested on a retired finite-size entry and is withdrawn in Paper 34. These are MEASURED decompositions (the H Lamb autopsy closes to +1.72 MHz of the measurement; the LS-6a one-loop residual of −5.65 MHz is $\sim 70\%$ itemized); the autopsy makes the projection-chain dictionary's reach visible at the observable level.

*b. Convention exposures versus kernel gaps.* The program's signature finding-type is the *literature convention exposure*: a multi-observable global fit surfaces itemization differences between independent precision compilations (Eides, Karshenboim, Krauth, Pachucki–Yerokhin, Antognini, Friar–Payne) at the sub-percent level the framework operates at. Nine exposures are catalogued (the later entries include architecture-class exposures beyond the convention class), including: the

$\sim$ 0.01%-of-$\nu_F$ recoil/multi-loop itemization difference that propagates to $\sim$ 25 mfm in an extracted Zemach radius; the Friar-moment convention difference between an RMS radius $\langle r^2\rangle^{1/2}$ and a first moment $\langle r\rangle$ (a three-dimensional Gaussian-profile factor $\langle r^2\rangle/\langle r\rangle^2 = 3\pi/8 \approx 1.178$, i.e. +17.7% on the Friar moment vs the Eides closed form); and the resolution of a $3\sigma$ helium fine-structure disagreement traced to a numerically incorrect relativistic Bethe-log itemization in an older compilation. The complementary finding-type is the *framework kernel gap*: places where the framework's leading-order operator (Eides Zemach, Foldy–Friar contact, Friar moment) is too coarse for the precision target. Each precision tightening tends to relocate the load-bearing systematic from one to the other; cataloguing which class a given observable lands in is itself the program's output.

*c. The structural prediction that replaced the naive one.* A falsifiable structural PREDICTION emerges from re-tagging the matches catalogue by the number of Layer-2 inputs a chain consumes. An earlier form scaled the residual with projection *depth*; the May 2026 synthesis pass falsified that form on the catalogue itself (depth-3 chains exist at residual 0 and at $-211$ ppm; depth-4 chains at +2 ppm and at $-0.534$%). The replacement is the *Layer-2-presence bound*: the residual of a catalogue row is bounded above by the larger of the framework's basis-quality residual at zero Layer-2 inputs and the largest single Layer-2 input consumed,

$$|\epsilon(\text{obs})| \;\le\; \max\!\big(\epsilon_{\text{basis}},\; \max_i |L_2^{(i)}|\big).$$

Zero-Layer-2 chains run from machine-exact (Stefan–Boltzmann $\pi^2/90$, hydrogen static polarizability $\frac{9}{2}a_0^3$, $S^3$ Casimir 1/240) to basis-quality ceilings ($-0.534$% for the one-loop Lamb-shift closure); one-input chains tighten to tens of ppm or below; multi-input chains sit in the multi-focal-wall regime, dominated by the largest single input. This is a PREDICTION, consistent with the present catalogue with one known exception — the corrected D 1S HFS row, whose $-210.9$ ppm residual against a single $r_Z(D)$ Layer-2 input is an open case rather than support (Paper 34 §V.C) — and falsifiable by future calculation, not a phenomenological fit.

## VII. HONEST SCOPE AND POSITIONING

*a. What is robust.* The load-bearing results are structural and verified at their stated tiers. The $\pi$-free Layer 1 certificate is SYMBOLIC (exact arithmetic, both Coulomb $S^3$ and Bargmann–Segal $S^5$). The Klein–Gordon $\pi$-injection at temporal compactification is a SYMBOLIC panel (200 cases; $E^{S^3}_{\text{Cas}} = 1/240$ and the conformal-scalar chain exact). The one-body entanglement-inertness is an INTERNAL THEOREM with machine-precision numerical confirmation (the two-fermion HO zero-entropy rigidity that once stood beside it is retracted, v5.0.0). The two-layer thesis and the single-projection origin of physical content are stated as OBSERVATIONS, asserted at that tier and no higher. The projection dictionary is a concrete, structured artifact: every transcendental in the arc wears its projection tag.

*b. What is conditional.* The precision matches are sub-percent *framework-native* agreements, not first-principles predictions of the physical constants. Each consumes external Layer-2 inputs (Zemach radii, $g$-factors, multi-loop QED coefficients, nuclear polarizabilities) in their native compilation conventions, and the program's value is precisely that it surfaces where those conventions and the framework's leading-order kernels are too coarse—a diagnostic instrument for precision systematics, not a replacement for the precision compilations. The block-entropy exponent $\gamma_\infty \approx 1.93$–$1.96$ is a MEASURED extrapolation, not the exact second-order value 2. The entropy area-law and the $R$-independence of composed block entanglement are properties of the construction, stated with their limitations, not physical statements about molecular entanglement at dissociation.

*c. The active wall.* The arc's principal open boundary is the *multi-focal-composition wall*: the framework couples discrete labels cleanly but has *no* native composition theorem for combining multiple Fock-style projections. Six independent instances are on record across domains (the Yukawa non-selection theorem, the LS-8a renormalization scale, the HF spatial recoil/Zemach/multi-loop attempts, the chemistry W1e wall). This is why multi-Layer-2-input observables sit in the residual-dominated regime of the Layer-2-presence bound, and why the precision program's deepest residuals are structural rather than numerical. The wall is named as an active boundary, not papered over.

*d. Positioning.* The natural reading of this arc is as a *vocabulary and a diagnostic*. The two-layer framework gives every GeoVac observable a definite answer to "where does each transcendental enter," and the multi-observable focal-length program turns that vocabulary into an instrument that exposes convention mismatches and kernel gaps in precision atomic physics. The discrete skeleton and its continuous projections are dual descriptions; the contribution is structural traceability, not an ontological claim and not a derivation of the constants the projections consume.

[1] J. Loutey, "Angular Momentum as Information Geometry: Isotropic Packing and the Recurring Angular Cross-Section," GeoVac Paper 0 (2026).

[2] J. Loutey, *The Fine Structure Constant as a Spectral Coincidence on the Hopf Fibration* (GeoVac Paper 2, observation status), Zenodo (2026).

[3] J. Loutey, "The Dimensionless Vacuum: Recovering the Schrödinger Equation on a Scale-Invariant Label Lattice," GeoVac Paper 7 (2026).

[4] J. Loutey, "Spectral-Geometric Exchange Constants: Projecting Discrete Spectra onto Continuous Manifolds," GeoVac Paper 18 (2026). [Master Mellin engine, §III.7; transcendental-tier taxonomy.]

[5] J. Loutey, "The Bargmann–Segal Lattice: $\pi$-Free Discretization of the Harmonic Oscillator on $S^5$," GeoVac Paper 24 (2026). [Layer-1 $\pi$-free certificate; HO rigidity theorem.]

[6] J. Loutey, "Entanglement Structure of the Angular Momentum Eigenbasis: Energy-Entanglement Decoupling and Basis-Intrinsic Sparsity," GeoVac Paper 26 (2026).

[7] J. Loutey, "Entropy as a Projection Artifact: One-Body Operators are Entanglement-Inert on Sparse Lattices," GeoVac Paper 27 (2026).

[8] J. Loutey, "QED on $S^3$: Spectral Geometry of Perturbative Transcendentals," GeoVac Paper 28 (2026). [Graph-native QED $\pi$-free certificate.]

[9] J. Loutey, "The Universal/Coulomb Partition of the GeoVac Spectral Triple," GeoVac Paper 31 (2026).

[10] J. Loutey, "The GeoVac Spectral Triple: A Synthesis Paper Locking the Framework into the Marcolli–van Suijlekom Gauge-Network Lineage," GeoVac Paper 32 (2026).

[11] J. Loutey, "GeoVac as a Two-Layer Framework: Projection Taxonomy and the Empirical Matches Catalog," GeoVac Paper 34 (2026).

[12] J. Loutey, "Time as Projection: The Klein–Gordon Spectrum on $S^3 \times \mathbb{R}$ and the Algebraic / Observation Split," GeoVac Paper 35 (2026).